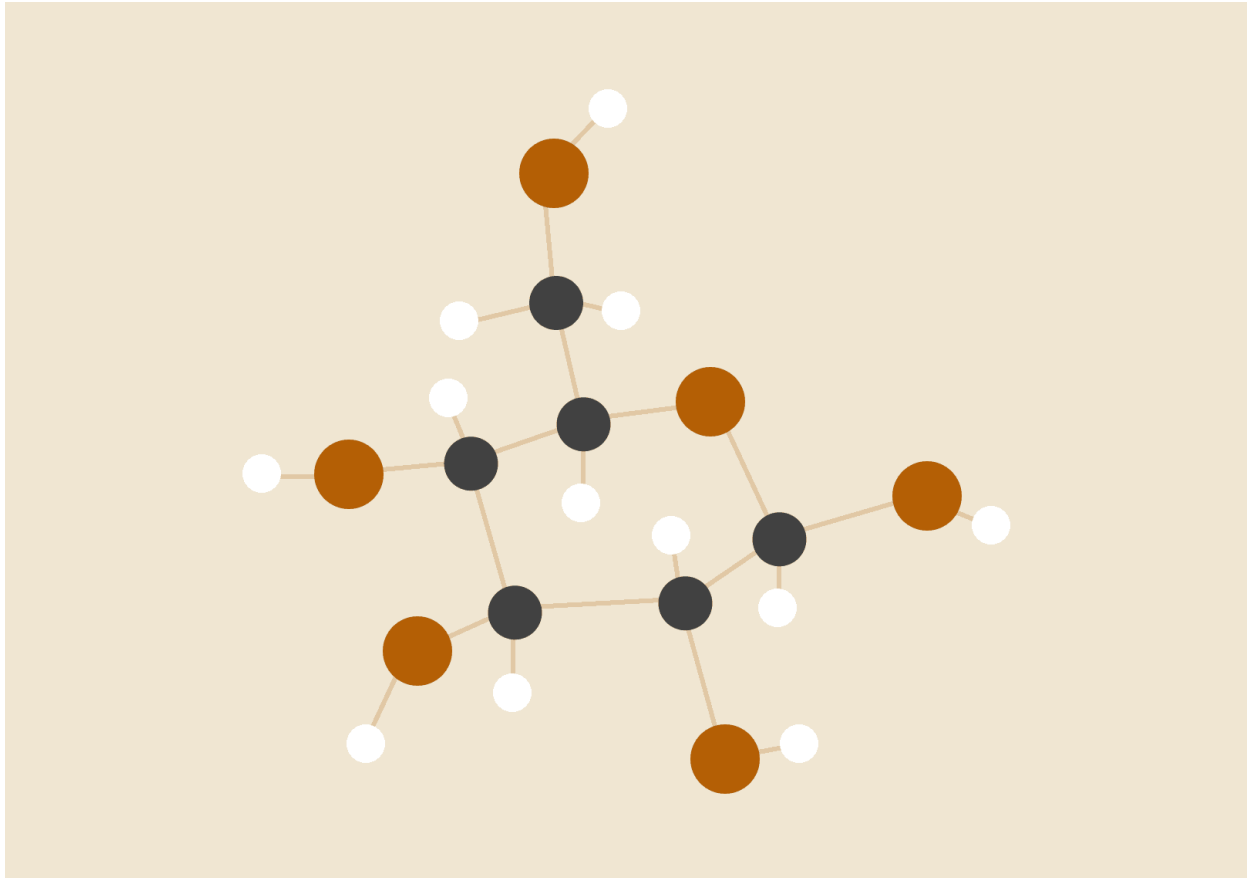


# Rambal Internship

*My experience at a 2-week internship and immersion program at Rambal Pvt. Ltd.*



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## INTRODUCTION

In this report, I will be delving into my experience at a 2-week internship and immersion program at Rambal Private Limited, a manufacturing firm based out of Thiruporur, Tamil Nadu, India. They are experts in the field of auto parts and specialise in the production of oil and air-based suspension systems for light commercial vehicles. Their vast and varied product encompasses over 10 products such as water pumps and steering systems. Throughout my internship, I was exposed to extensive knowledge about how a company of this scale and size operates on a day-to-day basis.

## DAY 1

On my first day, I was given a general overview of the Thiruporur production plant. First, I observed the production plant, where they received raw materials like rods, tubes and plates. The raw materials were then subjected to stringent quality tests and then went through the processes required to make them suitable for assembly. In particular, I focussed on the production of a piston rod, perhaps the most crucial element of a suspension system. I was handed a “process flow” document, detailing all the key processes such as seam welding, grinding and milling that would transform ordinary raw materials into a finished part ready for assembly. What truly amazed me was the minuscule margin for error of around  $\pm 0.1\text{mm}$ , and how accurate and specific the product drawings were. I also was introduced to gas springs, which are used to dampen and control motions such as shutting boots and closing doors/windows. Finally, I was delighted to see the modelling and CAM software that were used to manufacture the products. I explored SolidWorks on a basic level and was able to make some connections with the AutoCad I had been exposed to in my SISC science program. Moreover, I was able to connect the dots with my Design and Technology schoolwork - I was able to make sense of most of the technical drawings and the annotations that accompanied them.

## DAY 2

On my second day, I was familiarised with air suspension. This was truly an exciting moment for me because, for the first time in my life, I got the opportunity to observe the undercarriage of a vehicle. I could barely contain my enthusiasm as I saw the steering arm, transmission shaft, propellor shaft, and suspension itself. I was taken through the

flow of the compressed air, from a compressor to through a series of valves and reservoir tanks and finally to the bellows that would be used in the air suspension system to cushion the bumps. What truly astonished me was how ingenious the solutions to problems were. For example, the compressor needed a power source to work. Using the vehicle's battery would be dangerous as that is one of the most replaced parts on a vehicle. To combat this, they used the keyline as a power source and some relays to step up the power, yet another thing I had been previously acquainted with as a result of my Arduino Intermediate Course. After lunch, I was shown how a Vertical Machining Centre (VMC) works. Here, I learnt the difference between MCode (Machine Code) and GCode(Geometric Code). I understood the code line by line and saw how the structure of the code itself was not too different from other coding languages that I had been previously exposed to - a tool was declared similar to a variable, and the series of movements that the tool needed to perform were called similar to functions. To conclude the day, to my excitement, I got to operate one of the machines myself using the dial and saw how the tool operated using x,y and z commands.

## DAY 3

On my third day, I delved deeply into the assembly process. This experience was incredibly enlightening for me, as it was the first time I got to see the intricate steps involved in putting together various components of the GEN4 front and rear damper assemblies. I observed each step of the assembly with detail, taking pictures and fully understanding what was happening at each step as the piston rod, rod guide, oil filling, strut attachment and knuckle/wheel hub setup all came into place. Later, I made some drawings pertaining to what I had observed, using sketching techniques I had learnt in my Design and Technology classes. To conclude the assembly, I put together an entire end blank assembly multiple times.

Additionally, I had the chance to learn about Faraday's Law and its applications in the motors used in compressors. I was captivated by the simplicity yet enormous application of this principle. The theory itself was new to me, and seeing it used in real life at a massive scale for something crucial in everyday life, was truly eye-opening for me. I understood how these motors were essential for the compressors' functionality, converting electrical energy into mechanical work to

allow for an increase in the volume of air.

Another highlight of the day was learning about retarders. I discovered how these devices, often used in heavy vehicles, help in slowing down the vehicle by dissipating kinetic energy and reducing the reliance on traditional friction brakes. This not only improves safety but also prolongs the lifespan of brake systems, and is especially crucial while descending from inclined surfaces.

## DAY 4

On the fourth day, I moved from production to the office upstairs to learn about the organisational structure of the company, including the Finances, Enterprise Resource Planning, Product Design, Research and Development, and many more. First up was finance. I acquired knowledge of the general finances of a company, including balance sheets, profit and loss documents as well as cash flow statements. Each document provided a different perspective concerning the financial health of the company, and I was fascinated to see how they all fit together like pieces in a jigsaw puzzle. I was also introduced to bookkeeping and accounting for the first time and saw how each and every transaction is masterfully captured in a ledger. I then proceeded to go through some of the most recent invoices and Goods Receipt Notes in the company and was able to glean large amounts of information pertaining to HSN codes, Central GST and State GST (CGST and SGST). I grasped the knowledge of the daily expenses that the company incurred, from subcontractors to suppliers. Later that day, I became well-versed in the company's weekly and monthly transactions with the bank. I learnt about how Rambal, as an export-based enterprise and an MSME, was eligible for certain types of financial services and tax refunds that were inaccessible to the general public. These included pre and post-shipment loans, which provided liquid cash for day-to-day operations and supplier payments before the customer payment was received.

## DAY 5

On the fifth day, I was given the opportunity to explore the ERP system in-depth. The entire process of receiving and processing a customer PO, quoting a price, designing, purchasing raw materials, maintaining stock, production, subcontracting, and shipment

was explained to me in depth. After this, I saw how the homegrown ERP system encompassed these fields and compartmentalised them into easily accessible data. This was perhaps one of the only flaws I had seen so far, as the software used for ERP was 20+ years old and in severe need of an update. Even still, it was fascinating to see how the system interconnected the workers to the stores, the stores to the purchasing team and the purchasing team to the management. Thanks to this flow of information, data such as the stock of a certain part, the number of finished goods etc. was available to every department. People were able to make informed decisions about what materials to buy and where the materials were.

## DAY 6

On the sixth day of my internship, I met with the head of the aluminium casting department. He was an experienced foundryman, working for more than 30 years with leading companies. He was in charge of setting up the new aluminium casting division in Rambal and had full control over the production. I learnt about how castings are made, the different processes involved, and how they are perhaps die-casting is one of the single most important processes in the entire manufacturing industry itself due to its enormously widespread uses. Next, I was unexpectedly called to participate in an impromptu meeting of all the heads of the departments, where the Managing Director of the company described his experience at Roots Group in Coimbatore, one of India's leading exporters. There was then a discussion about how they could incorporate some of the processes and learnings at Roots into Rambal to increase the efficiency and quality of their production, such as integrating IoT into assembly processes. To be a part of this meeting was truly a gratifying experience for me, as I felt I was a part of the company and not just there for an internship as I saw how they functioned on a day-to-day basis. After that, I saw the existing machines used for the production of castings and understood their significance in the die-casting process.

## DAY 7

The seventh day of the internship was the most captivating to me as it aligned with my interests the most. I first got a chance to see what softwares they used on a daily basis. These included AutoCAD, SolidWorks, Catia, Ansys, and many more. I was most riveted

by Ansys Workbench, an engineering simulation software that produced accurate data about the strength and feasibility of the components and their materials - something I had never seen nor heard of before. I then engaged myself in an experiment using an inertial sensor to test the dampening forces of suspension and how they affected the pitch and roll of a vehicle on a bumpy road. Needless to say, this was fascinating, as I could participate in an experiment that would benefit the Research and Development department.

## DAY 8

On the eighth day, I explored the last two departments left; two that probably interacted most with each other on a day-to-day basis - Purchase and Stores. I explored how suppliers were chosen, audited, and negotiated with. I observed how the process of commodity purchase occurred and how the Bill of Materials was used along with the standard rates for raw materials to manipulate the price of the output. One very interesting learning experience for me was to get a feel of the London Metal Exchange and how they determined standard rates for most metals used in the manufacturing process. Later that day, I went down to the stores to see how they integrated counting machines and ERP systems to maintain a stock of all the raw materials and “boughtouts” that were processed by the company. The most interesting and perhaps ingenious part of the day was when I saw how oil and coolant were extracted from the burr scrap and then packaged and sold to make a quick penny.

## DAY 9

On the last day of my internship, I was given the enormously significant opportunity to give a tour of the entire factory, its production facilities, product lineup, and processes to a potential investor. My two-odd weeks of learning culminated in this moment where I was able to answer more than 90% of the questions posed by the investor with little to no hesitation, backed up with accurate facts and figures.